

# Vignesh Jagathese

*Visiting Assistant Professor, NDSU*

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I am a US Citizen.

## Academic Appointments

2026-Present **Visiting Assistant Professor, North Dakota State University (NDSU)**  
Postdoctoral Mentor: Janet Page

## Education

May 2026 **PhD, Mathematics, University of Illinois Chicago (UIC)**  
*Permanence Properties of Quasi- $F$ -Singularities over Arbitrary Fields*, Advised By Kevin Tucker  
April 2020 **Masters of Science, Mathematics, University of Michigan (UofM)**  
April 2020 **Bachelor of Science, Honors Mathematics, University of Michigan (UofM)**

## Mathematical Interests

Commutative Algebra, Algebraic Geometry,  $F$ -Singularities, Quasi- $F$ -Splittings,  $F$ -Pure and Plus-Pure Thresholds, Singularities via Cartier Operators

## Work Experience

2020-2021 **VECO Power Trading, Quantitative Trader and Researcher**  
Developed models to predict day ahead energy prices, written using R, PHP, MySQL and Python.  
Summer 2019 **Chicago Trading Company, Quantitative Trading Intern**  
2017-2018 **University of Michigan Solar Car Team, Operations Director**

## Publications and Preprints

- [3] **Bounds on the plus-pure thresholds of some hypersurfaces in (ramified) regular rings**, (2026), with Marta Benozzo, Vaibhav Pandey, Pedro Ramirez Moreno, Karl Schwede, and Prashanth Sridhar, 2509.07217, (*Mathematische Zeitschrift* 313 (2). Article no. 32)
- [2] **On quasi- $F$ -purity of excellent rings**, (2026), 2503.10778, (*Communications in Algebra* 54 (4). 1760-1771)
- [1] **On  $F$ -pure thresholds and quasi- $F$ -purity of hypersurfaces**, (2025), with Jack J Garzella, 2509.06211, Submitted

## Awards and Grants

January 2027 **Collaborate@ICERM**, Funded stay at ICERM as member of a collaboration investigating plus-pure thresholds.  
2025 **UIC Award for Graduate Research**, The Graduate College's longest-running support for research by graduate students at UIC. Comes with cash stipend  
2023 **UIC Professor Herbert Alexander Award**, Departmental award, with cash stipend  
2022-2026 **NSF RTG Fellowship**, Research and Training (pre-doctoral) Grant  
Spring 2019 **NSF REU**, Research Experience for Undergraduates administered through UofM

## Teaching

### Instructor of Record Positions

- Fall 2026 **MATH 144**, *Mathematics For Business*, NDSU
- Summer 2026 **SEMW 108**, *Summer Enrichment Math Workshop—Intermediate Algebra*, UIC
- Spring 2025 **MATH 109**, *College Algebra Workshop*, UIC
- Fall 2023 **MATH 108**, *Intermediate Algebra*, UIC
- Spring 2023 **MATH 104**, *Mathematical Reasoning Workshop*, UIC
- Fall 2022 **MATH 109**, *College Algebra Workshop*, UIC

### Administrative Positions

- Fall 2025 **Teaching Assistant Coordinator**, UIC

I worked with the Associate Head of Instruction to train and evaluate new TAs in the department.

### Teaching Assistant Positions

- Spring 2025 **MATH 110**, *College Algebra*, UIC
- Spring 2023 **MATH 105**, *Mathematical Reasoning*, UIC
- Fall 2022 **MATH 110**, *College Algebra*, UIC
- Fall 2021 **MATH 110**, *College Algebra*, UIC
- Fall 2021 **MATH 125**, *Elementary Linear Algebra for Business*, UIC

### Grading Positions

- Fall 2023 **MATH 516**, *A Second Course in Abstract Algebra I*, UIC
- Winter 2020 **MATH 417**, *Matrix Algebra*, UofM
- Winter 2019 **MATH 452**, *Multivariable Analysis*, UofM

## Invited Talks

- 07/2026 **UIC Young Scholars Program**
- 04/2026 **AMS Sectional at North Dakota State University: Topics in Commutative Algebra**
- 02/2026 **University of Utah Commutative Algebra Seminar**
- 01/2026 **JMM special session: Advances in Commutative Algebra**
- 11/2025 **University of Missouri Algebra Seminar**
- 10/2025 **University of Nebraska Commutative Algebra Seminar**
- 09/2025 **Purdue Commutative Algebra Seminar**
- 05/2025 **UC San Diego Algebraic Geometry Seminar**
- 03/2024 **UIC Incoming Students Colloquium (Also 03/2025)**
- 01/2023 **UIC Graduate Student Colloquium (Also 04/2024, 01/2025)**
- 4/2022 **UIC Undergraduate Math Club (Also 08/2023)**

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## Service

### Journal Referee

Refereed for Mathematische Zeitschrift.

#### 2024-2025 **UIC Directed Reading Program Advisor**

- **(Spring 2025)** Advised statistics undergraduate Juan Garcia on introductory abstract algebra.
- **(Fall 2024)** Advised electrical engineering and physics undergraduate Caleb Williams on applications of category theory.
- **(Spring 2024)** Advised mathematics undergraduate Max Nguyen on sheaf cohomology, specifically on computations via the Čech complex.

#### 2023-2024 **UIC Commutative and Homological Algebra Graduate Seminar Organizer**

Founded and organized a graduate seminar on active research topics in algebra.

#### 2022-2023 **UIC Mathematics Graduate Student Association Secretary**

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## Outreach

#### 2022-2026 **Math Circles of Chicago Teacher**

Taught weekend sessions and was an instructor for their summer programs in 2023 and 2026.

#### 11/2024 **Sonia Kowalevsky Math Day Organizer**

Helped organize an annual outreach program to help spark young womens' interest in math.

#### Summer 2024 **UIC Young Scholars Program Instructor**

Taught and helped organize a free summer program for Chicago high school students at UIC.

#### 2018-2020 **Wolverine Pathways Instructor**

#### 4/2019 **MMaths Math Competition Organizer**